



INTEGRATED PLATFORM FOR MONITORING CREW UTILISATION AND OPERATIONAL DISTRIBUTION

End-to-End Dashboard Platform for Macro-Level Monitoring of Crew Deployment, Operational Duty Metrics, Bio-Identifiers, Risk Flags, and Resource Utilization Across Zones and Divisions Using Data Visualization.

Number of distinct divisions	Total Unique Crew ID
2	4179
Designation Count	Traction Types
14	4
Total KMs Run	Total Trips
66066K	2,89,872

Developed & Visualized by:
Vaibhav Porwal
B.Tech. (Computer Science and Engineering)
USICT, Gautam Buddha University, Greater Noida
Gautam Budh Nagar (Uttar Pradesh) - 201312
Email : vp.porwal11@gmail.com

Total Duty Hours	Total Employee ID (Crew Strength)
169M	4,017
Active No. of Days	Total Shunt Count
109	164
Inactive Crew Count	Distinct Crew Base ID Count
4,079	4,187

Home

Crew Biodata & Distribution Overview

Distribution and Deployment Metrics

Operational Efficiency & Utilization Trends

Crew Risk & Operational Efficiency Patterns

Operational Trends & Comparative Insights

Crew Effort Analysis

Leave & Traction Utilization

Select Division

BSP

TDL

Select Date Range

31-03-2025

09-06-2025

Select Designation(s)

All

Total Crew

4179

Total Trips

289872

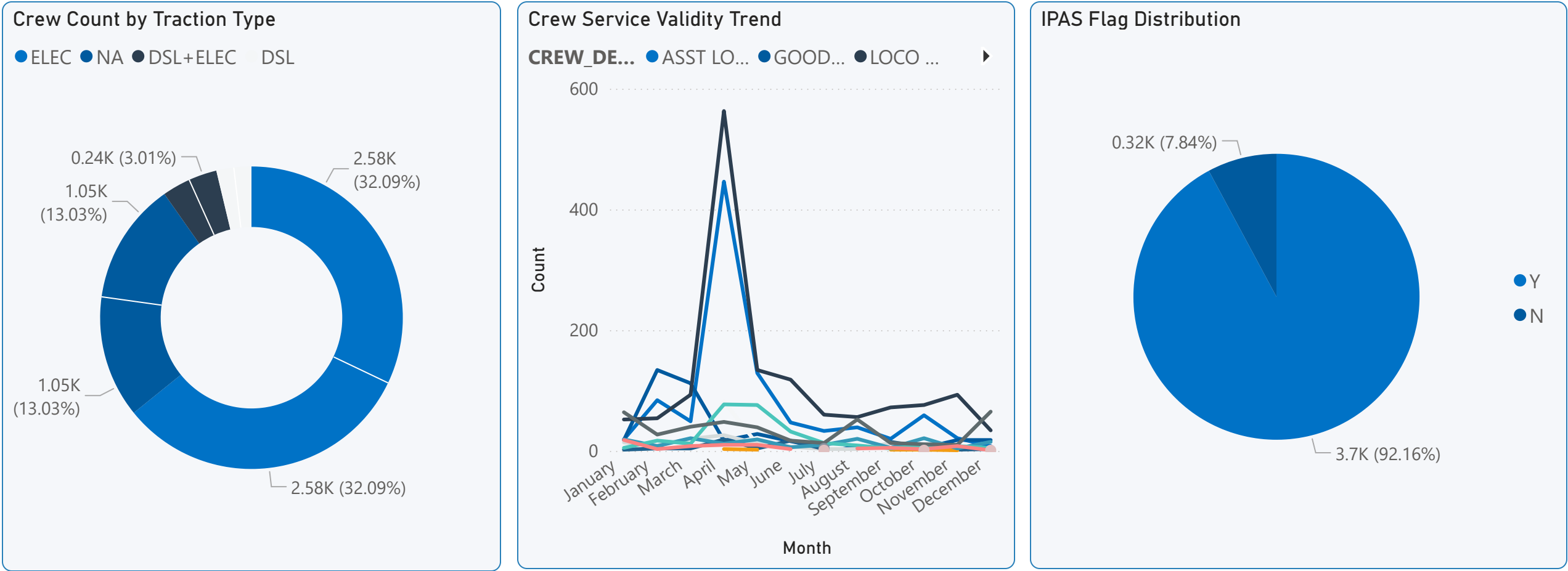
Total KMs Run

66065550

Avg KM per Trip

238

I. Crew Biodata & Distribution Overview



Select Division

BSPTDL

Select Date Range

31-03-202509-06-2025

Select Designation(s)

All

Crew ID Count

4179

Designations

14

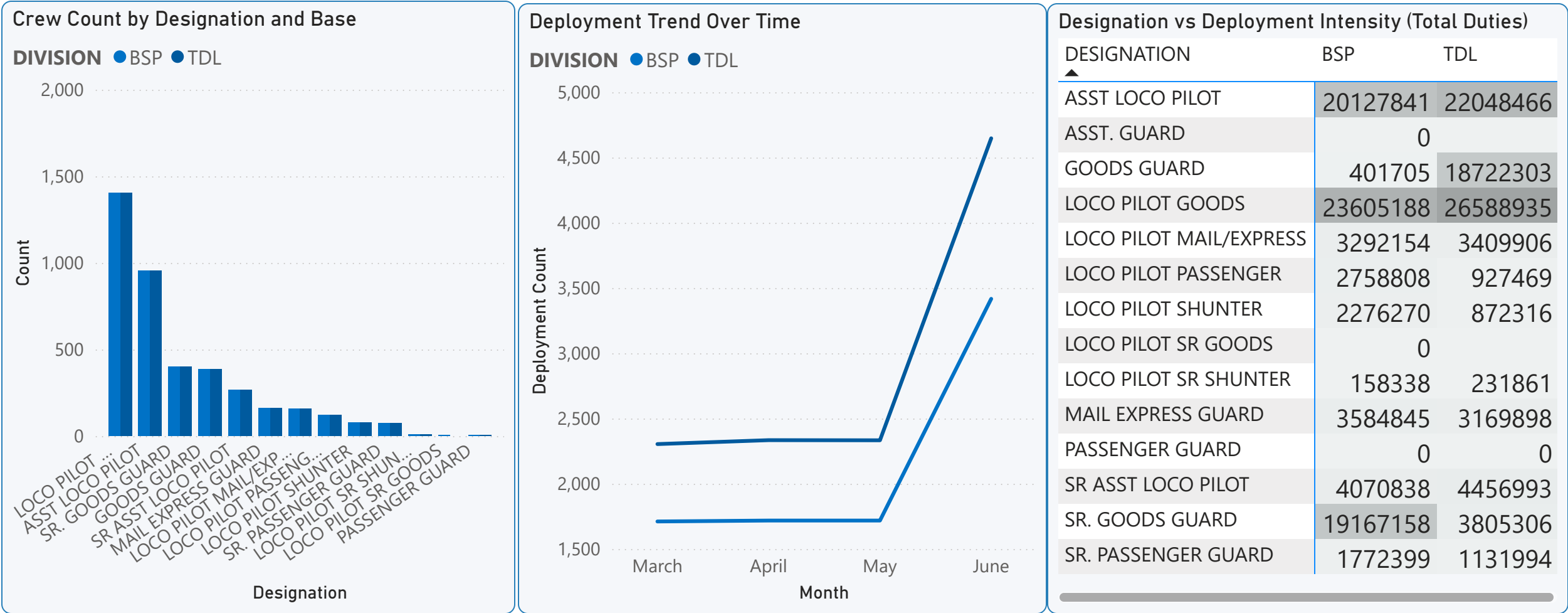
Active Deployment Days

109

Avg KM per Trip

238

II. Crew Distribution & Deployment Metrics



Select Division

BSP

TDL

Select Date Range

31-03-2025

09-06-2025

Select ORG TYPE(s)

All

Total Trips Made

289872

Designations

14

Total Duty Hours

169002213

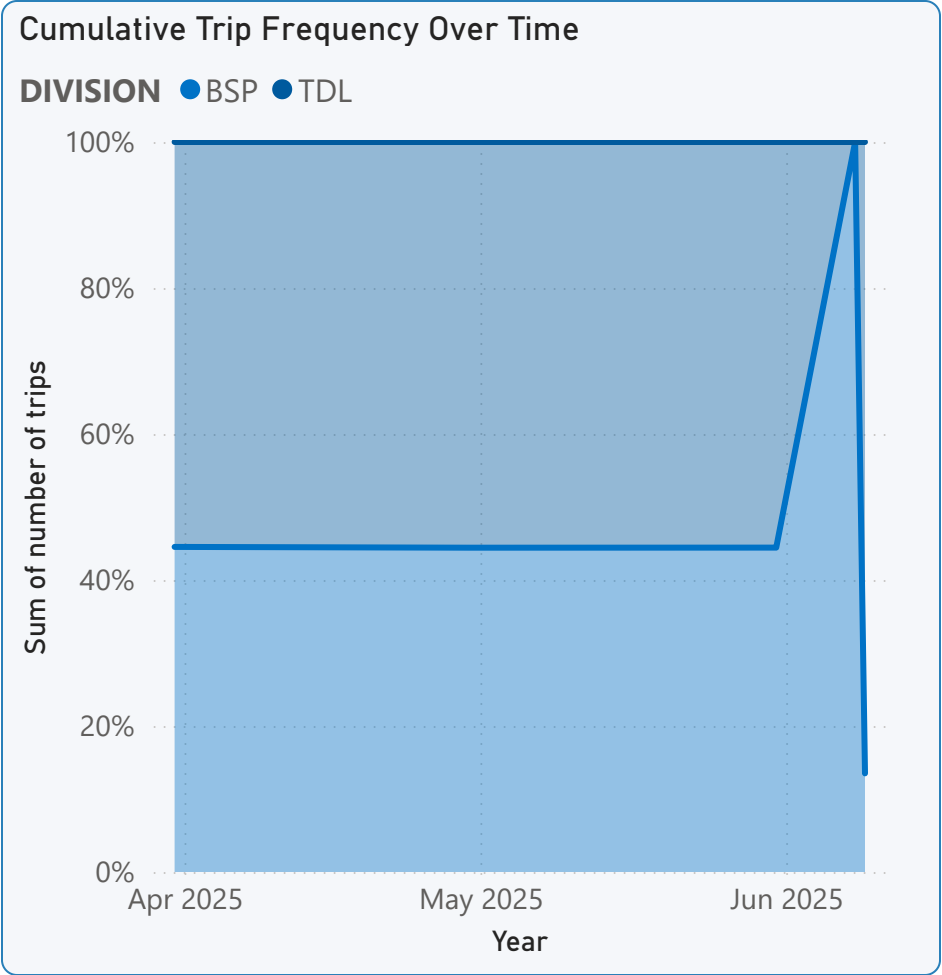
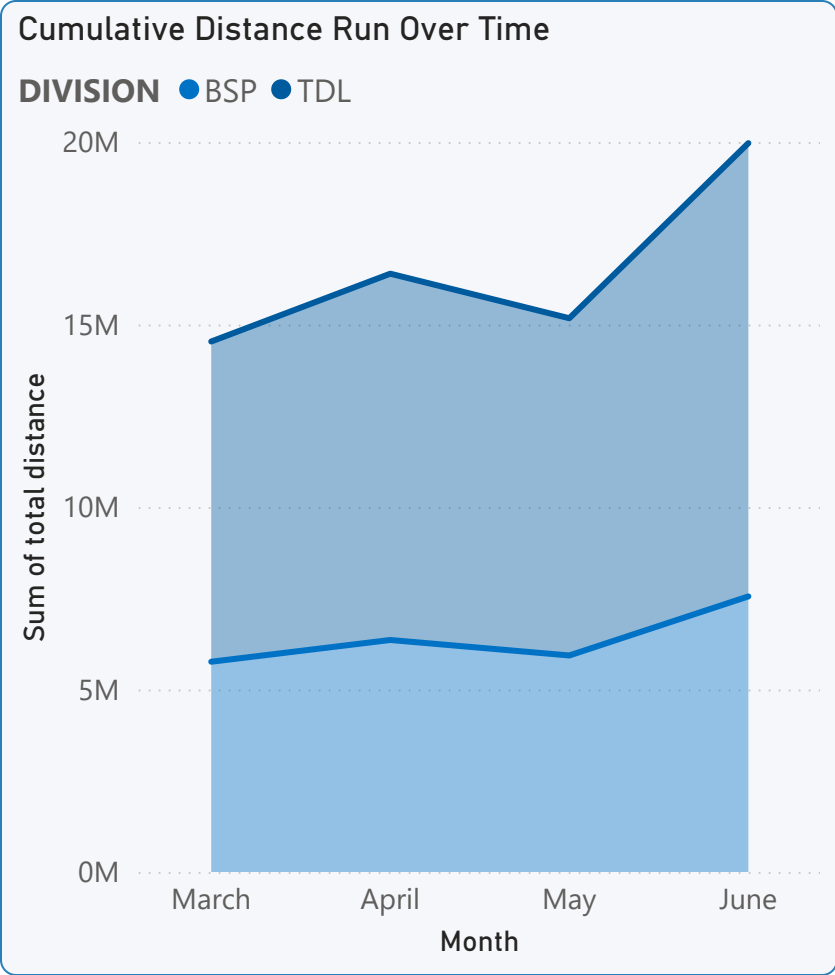
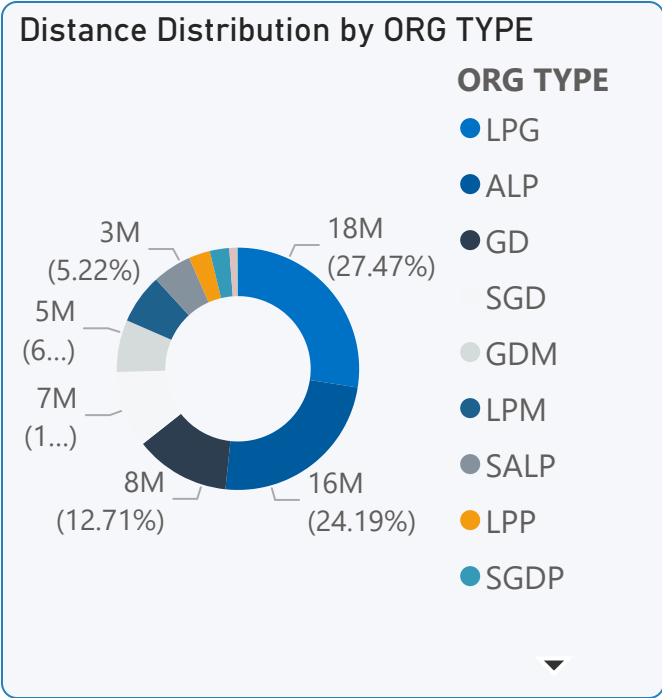
Avg Duty Hours per Trip

238

III. Operational Efficiency & Utilization Trends

Division-wise Trip Frequency

DIVISION	March	April	May	June
BSP	28660	31520	29599	38548
TDL	35699	39429	37009	49408



Select Division

BSP

TDL

Select Date Range

31-03-2025

09-06-2025

Select Designation(s)

All

Alcohol Cases Count

4074

RRA Total

305

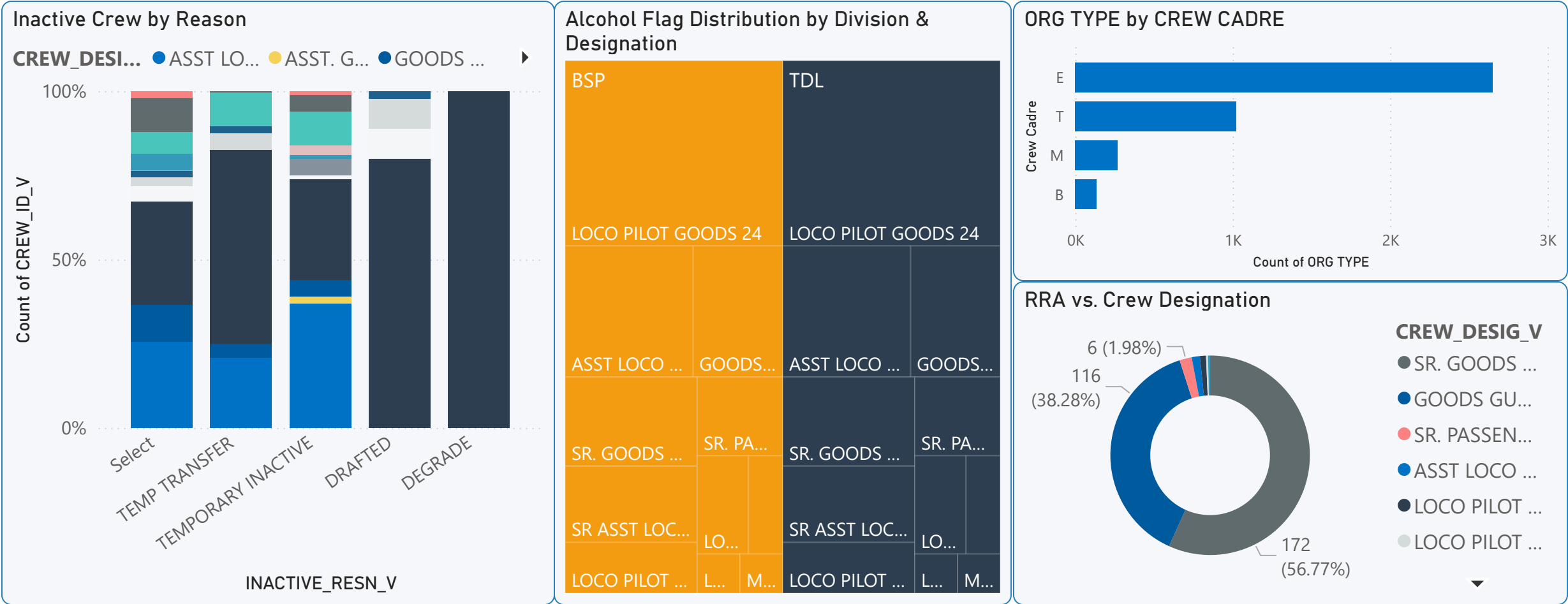
Inactive Crew Count

4,079

L Flag Count

4,079

IV. Crew Risk & Operational Efficiency Patterns



Select Division

BSP TDL

Select Date Range

31-03-2025 09-06-2025

Select Designation(s)

All

OSRA KMs

114K

Total Shunt Count

164

Avg Slot Number Used

168

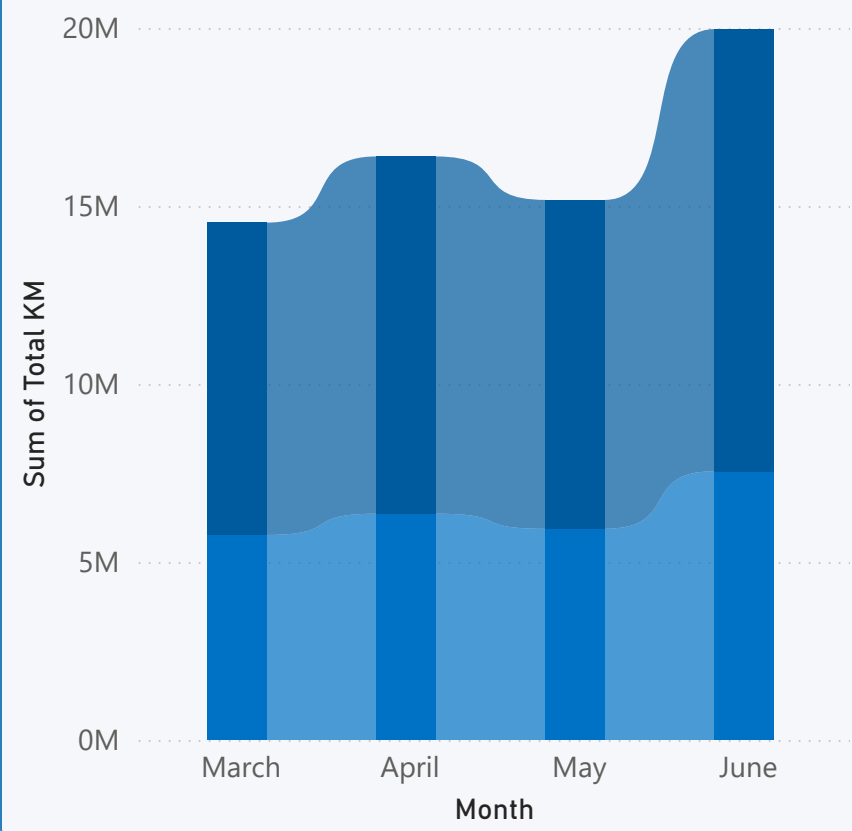
TEST_TRNG Sum

13887

V. Operational Trends & Comparative Insights

Monthly Ranking of Crew Bases by Total Kilometers

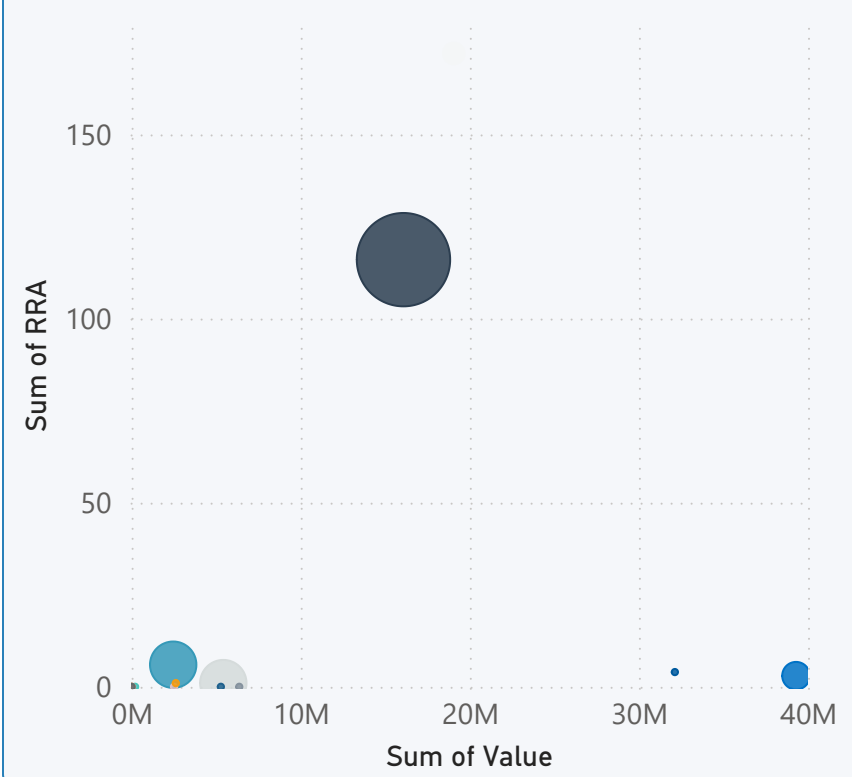
DIVISION ●BSP ●TDL



Duty Effort vs. Rest Allocation Analysis

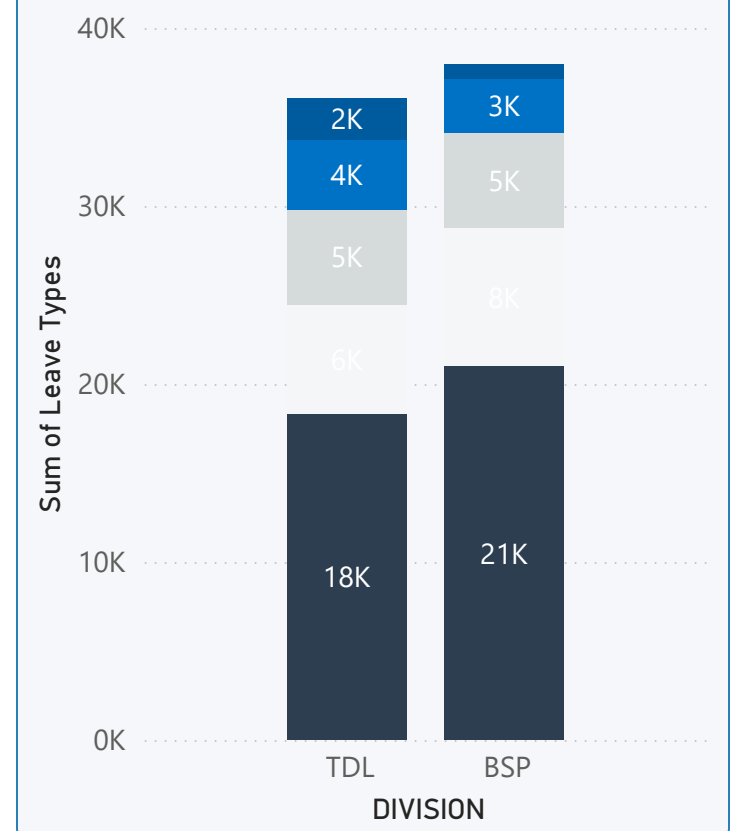
Correlates running duty effort with rest & burden

ORG TYPE ●AGR ●ALP ●GD ●GDM ●GDP ●LPG



Leave Type Distribution by Division

●SICK L... ●ABSENT ●LEAVE ... TEST T...



Select Division

BSP

TDL

Select Date Range

31-03-2025

09-06-2025

Select Designation(s)

All

Non Run Duty MINS

36M

Run Duty Mins

133410065

NGHT DUTIES (Mins)

44542829

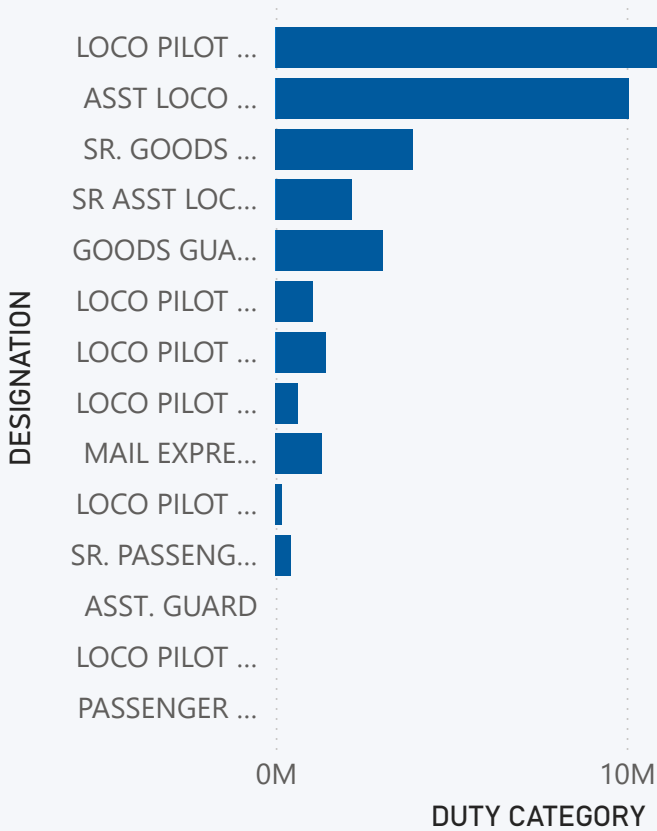
Avg Duty Per Crew

42,072

VI. Crew Effort Analysis

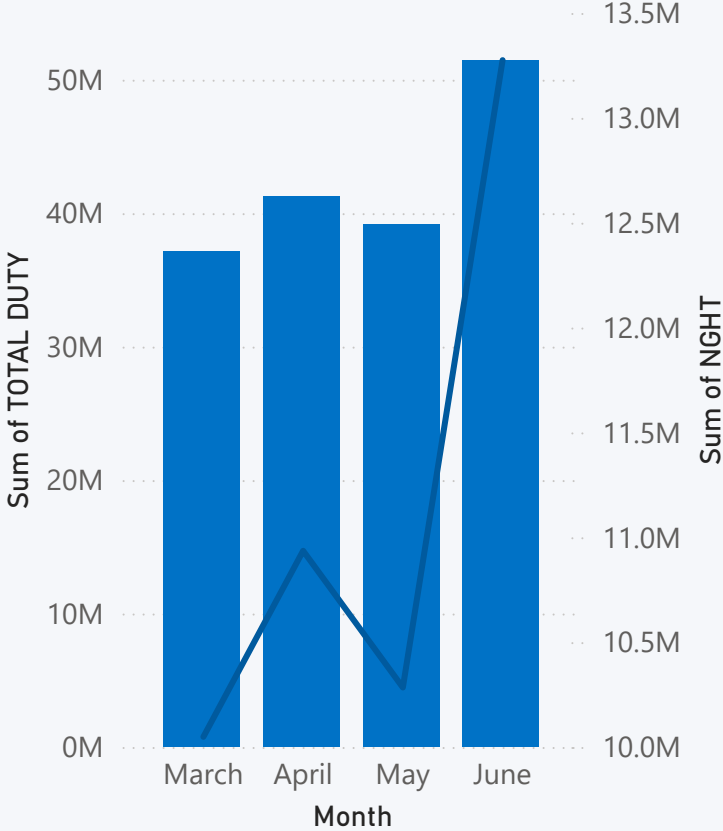
Duty Category Breakdown

● STATIONAY DUTY ● NON RUN DUTY MIN



Monthly Trends of Total Duty vs. Night Duty

● Sum of TOTAL DUTY ● Sum of NGHT



Average Crew-wise Coaching Duty

ORG TYPE	RUN_DUTY_MIN_N	FOOT_PLT_KM_N
AGR	0.00	0.00
ALP	1412.66	794.99
GD	121.07	50.64
GDM	6779.27	5,258.39
GDP	0.00	0.00
LPG	259.21	116.76
LPM	6723.82	4,649.72
LPP	4357.59	2,062.82
SALP	1762.81	1,192.38
SGD	1424.83	653.40
SGDP	6793.76	3,970.89
SHT	0.00	0.00
SLPG	0.00	0.00
SSHT	0.00	0.00
Total	1480.96	901.96

Select Division

BSP

TDL

Select Date Range

31-03-2025

09-06-2025

Select Designation(s)

All

Traction Types

4

Average Slot Allocation

167.87

Total ALKM Leave

31120

Total ALKM Non-Leave

9841

VII. Leave and Traction Utilization Analysis

